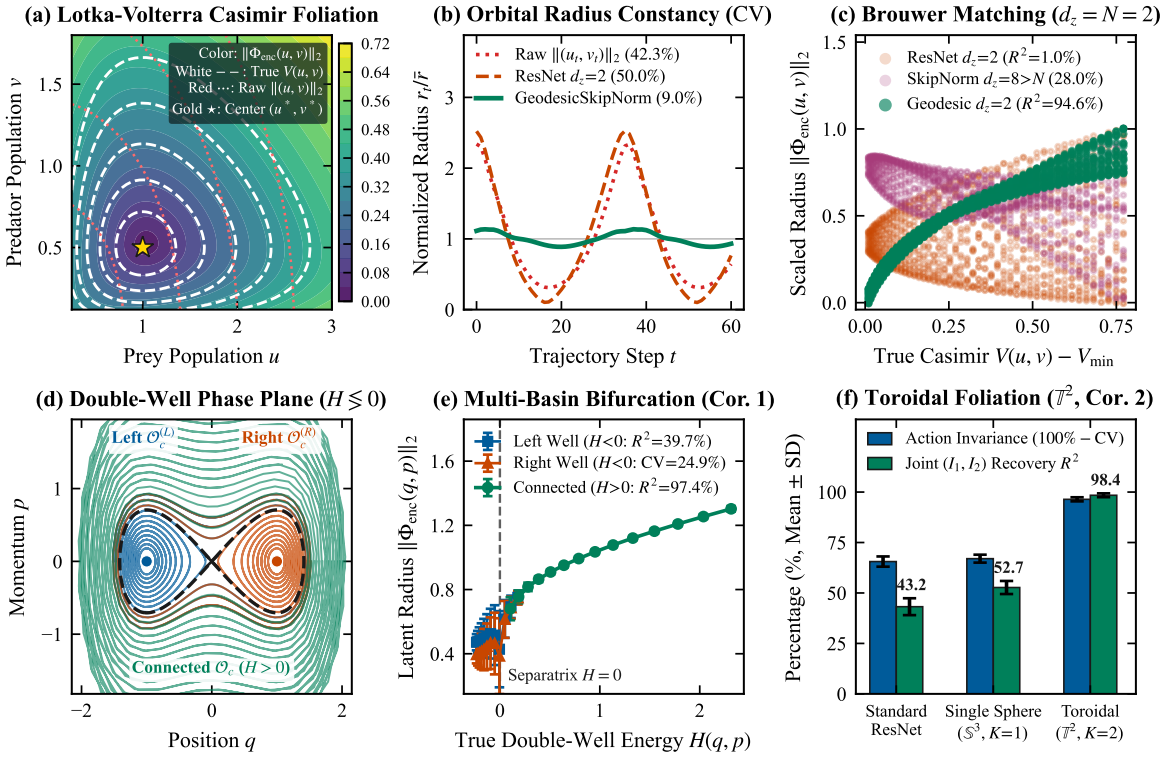

(a) Lotka-Volterra Casimir Foliation
Color: $\|\Phi_{\text{enc}}(u,v)\|_2$
White – – : True $V(u,v)$
Red ···: Raw $\|(u,v)\|_2$
Gold ★: Center (u^*, v^*)
Predator Population v
Prey Population u
(b) Orbital Radius Constancy (CV)
Raw $\|(u_t, v_t)\|_2$ (42.3%)
ResNet d_z=2 (50.0%)
GeodesicSkipNorm (9.0%)
Normalized Radius $r_t/\bar{r}$
Trajectory Step t
(c) Brouwer Matching ($d_z = N = 2$)
ResNet d_z=2 (R^2=1.0%)
SkipNorm d_z=8>N (28.0%)
Geodesic d_z=2 (R^2=94.6%)
Scaled Radius $\|\Phi_{\text{enc}}(u,v)\|_2$
True Casimir $V(u,v) - V_{\min}$
(d) Double-Well Phase Plane ($H \lesssim 0$)
Left $\mathcal{O}_c^{(L)}$
Right $\mathcal{O}_c^{(R)}$
Connected $\mathcal{O}_c$ ($H > 0$)
Momentum p
Position q
(e) Multi-Basin Bifurcation (Cor. 1)
Left Well (H<0: R^2=39.7%)
Right Well (H<0: CV=24.9%)
Connected (H>0: R^2=97.4%)
Separatrix $H = 0$
Latent Radius $\|\Phi_{\text{enc}}(q,p)\|_2$
True Double-Well Energy $H(q,p)$
(f) Toroidal Foliation ($\mathbb{T}^2$, Cor. 2)
Action Invariance (100% – CV)
Joint (I_1, I_2) Recovery R^2
43.2
52.7
98.4
Percentage (%, Mean ± SD)
Standard ResNet
Single Sphere ($\mathbb{S}^3$, K=1)
Toroidal ($\mathbb{T}^2$, K=2)